

Chapter 11 Index Laws

A convenient way of writing $2 \times 2 \times 2$ is

$$2^3 \quad \begin{array}{l} \leftarrow \text{Index} \\ \leftarrow \text{Base} \end{array}$$

Index Law 1

$$a^m \times a^n = a^{m+n}$$

Index Law 2

$$a^m \div a^n = a^{m-n}$$

Index Law 3

$$(a^m)^n = a^{m \times n}$$

Zero Index

$$a^0 = 1$$

Chapter 2 Integers

Integers are:

Positive whole numbers: 1, 2, 3, 4, 5, 6, ...

and Zero: 0

and Negative whole numbers: -1, -2, -3, -4, -5, -6, ...

Integers can be placed on a number line.

Integers on the right are larger than integers on the left.



Chapter 3 Algebra

★ $2a + 3a = 5a$, $4b - 6b = -2b$, $3c \times -2c^2 = -6c^3$, $18ad \div 3a = 6d$

★ Distributive law: $a(b+c) = ab + ac$

★ Factorising: $ab + ac = a(b + c)$

Chapter 4 Probability

★ Probability is the chance of something happening. The chance can be expressed as a decimal, fraction, percentage, ratio, or as words.

★ Theoretical probability = $\frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}}$

★ The complement of A is not A. $P(\text{not } A) = 1 - P(A)$